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Network Traffic Analysis of Parameter Tampering Attack Using Snort in a Web-Based Ordering System

Group 15

Names : 1. Azzahra Fatma Wahidah
2. Nazilatun Natasya Ramadhani

Class : 4CS2

Faculty : Ivan Firdaus S.T

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Name of Faculty : Ivan Firdaus S.T

Names of Developer :

1. Azzahra Fatma Wahidah
2. Nazilatun Natasya Ramadhani

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ABSTRACT

Web-based ordering systems are vulnerable to parameter tampering attacks due to weak server-side validation and improper access control mechanisms. In the previous project, a parameter tampering vulnerability was identified in the ARDWD (Around D' World) web-based ordering system, allowing attackers to manipulate HTTP POST request parameters using an interception proxy tool. This second project extends the previous study by focusing on network traffic analysis of parameter tampering attacks using the Snort Intrusion Detection System (IDS). The objective of this research is to analyze abnormal HTTP traffic generated during parameter tampering attacks and evaluate Snort's ability to detect suspicious network activity. The experiment is conducted in a controlled cybersecurity laboratory environment consisting of Kali Linux as the attacker machine, Cybersecurity Lab VM as the server and monitoring system, and the ARDWD web application as the target. Burp Suite is used to generate parameter tampering traffic, while Snort operates in IDS mode to monitor network packets and generate alerts. The results demonstrate that abnormal HTTP traffic generated during parameter tampering can be monitored and analyzed using Snort IDS, providing visibility into suspicious network behavior. This study highlights the importance of network traffic monitoring and intrusion detection systems in strengthening web application security and detecting malicious activity.

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METHODOLOGY

3.1 Testing Approach

This project focuses on analyzing network traffic generated by parameter tampering attacks using Snort Intrusion Detection System. The testing approach simulates an attacker who modifies HTTP POST parameters using Burp Suite and sends manipulated requests to the web server while Snort monitors network traffic in IDS mode.

The objective of this testing approach is to observe abnormal HTTP traffic patterns and analyze Snort alert detection during the attack process. The approach emphasizes network-level monitoring to evaluate how intrusion detection systems respond to suspicious parameter manipulation in web-based ordering systems.

3.2 Normal System Flow

The normal system flow of the ARDWD web-based ordering system follows a standard transaction process. Customers select menu items, fill in personal information such as name, phone number, and delivery address, and submit the order through the web interface. The HTTP request is then sent to the server, processed, and stored in the system database before displaying the order confirmation page.

During this process, Snort monitors HTTP traffic between the client and the server to observe normal network communication patterns. Under normal conditions, HTTP requests are transmitted without parameter modification and Snort records regular traffic without generating alerts.

METHODOLOGY

3.3 Interception-Based Attack Simulation

To generate abnormal network traffic, a parameter tampering attack is simulated using Burp Suite as an interception proxy. The attack modifies HTTP POST parameters before the request reaches the server in order to create suspicious traffic patterns that can be detected by Snort.

In this scenario, the attacker intercepts the order submission request, alters critical parameters such as customer name, address, quantity, or price, and forwards the modified request to the server. Snort monitors the manipulated traffic and generates alerts when abnormal HTTP requests are detected.

This simulation allows the system to produce parameter tampering traffic that can be analyzed through Snort logs and network monitoring results.

3.4 Snort Monitoring Process

Snort is activated in IDS mode in the Cybersecurity Lab VM to monitor network traffic generated during parameter tampering attacks. The system captures HTTP packets and applies detection rules to identify suspicious parameters in real time.

Snort is executed using the following command: `sudo snort -A console -q -c /etc/snort/snort.conf -i enp0s3`

Snort monitors incoming and outgoing HTTP traffic and analyzes packets based on predefined detection rules.

Example rule: `alert tcp any any -> any 80 (msg:"WARNING! PARAMETER TAMPERING DETECTED"; content:"HackerBase"; sid:1000001; rev:1;)`

When suspicious HTTP traffic is detected, Snort generates alerts containing source IP, destination IP, protocol information, and warning messages related to parameter tampering detection.

SYSTEM OVERVIEW

2.1 Application Description

The tested system is a simulated web-based food ordering application called ARDWD (Around D' World), which was previously analyzed in the first project for parameter tampering vulnerability. The application allows users to select menu items, enter personal information, and submit delivery orders through a web interface.

In this second project, the same application is used as the target system to generate network traffic during parameter tampering attacks. The focus of this project is not on discovering new vulnerabilities but on analyzing abnormal network traffic generated by parameter tampering attacks using the Snort Intrusion Detection System.

2.2 System Architecture

The experimental environment consists of three main components:

1. Kali Linux as attacker machine
2. Cybersecurity Lab VM as web server and Snort IDS monitoring system ARDWD web application as target system
3. Burp Suite is used to intercept and modify HTTP POST requests, while Snort monitors network traffic in IDS mode and generates alerts when suspicious parameters are detected.

4. Flow system:

Attacker → Burp Suite → Web Server → Snort Monitoring → Alert Detection

2.3 Stored Order Data

The ARDWD application processes several order parameters:

1. id_order
2. Customer_name
3. no_telpon
4. Address
5. Order_item

These parameters are transmitted through HTTP POST requests and become the main source of network traffic analyzed in this project.

VULNERABILITY SCENARIO

The ARDWD web application contains a parameter tampering vulnerability identified in the previous project due to weak server-side validation. Attackers can modify HTTP POST parameters during transmission using interception tools.

This vulnerability generates abnormal network traffic, which becomes the focus of analysis in this project. The system does not verify the integrity of submitted data, allowing manipulated parameters to pass through the network without detection at the application layer.

Therefore, network-level monitoring using Snort IDS is required to analyze suspicious traffic and detect anomalies.

ATTACK SCENARIO

The attack scenario demonstrates how parameter tampering generates abnormal HTTP traffic in the ARDWD web-based ordering system and how Snort Intrusion Detection System monitors suspicious network activity.

In this scenario, the attacker interacts with the ARDWD web application through the normal ordering process and submits a food order through the web interface. During the submission process, Burp Suite is used as an interception proxy to capture the outgoing HTTP POST request before it reaches the web server.

After intercepting the request, the attacker modifies critical parameters such as Customer_name, Address, or Order_item to generate manipulated HTTP traffic. The modified request is then forwarded to the web server, allowing abnormal parameters to pass through the network due to weak server-side validation in the application.

While the manipulated request is transmitted, Snort operates in IDS mode and monitors HTTP traffic between the attacker and the web server. Snort analyzes network packets and applies detection rules to identify suspicious parameter values in the HTTP POST request. When abnormal traffic is detected, Snort generates alert messages indicating parameter tampering activity.

This attack scenario produces abnormal HTTP network traffic that becomes the main focus of Snort monitoring and network traffic analysis in this project.

ATTACK ANALYSIS

5.1 Phase 1 – HTTP Request Interception

The Burp Suite intercepts outgoing HTTP POST requests and captures order parameters before reaching the server.

Captured parameters typically include customer name, delivery address, ordered items, quantity, and price information.

5.2 Phase 2 – Parameter Manipulation and Snort Monitoring

After intercepting the HTTP POST request, the attacker modifies critical parameters such as Customer_name or Address to generate abnormal network traffic. The manipulated request is then forwarded to the web server, allowing modified parameters to pass through the network.

During this process, Snort operates in IDS mode and monitors HTTP traffic in real time. Snort analyzes network packets and applies detection rules to identify suspicious parameter values within the HTTP POST request.

This phase demonstrates how parameter tampering generates abnormal HTTP traffic that can be observed and monitored at the network level.

```
[snort@secops ~]$ sudo snort -A console -d -u snort -g snort -c /etc/snort/local.rules -f snort
```

ATTACK ANALYSIS

5.3 Alert Output Analysis

Snort generates alert messages when suspicious HTTP traffic is detected. The alert output contains important network information such as source IP address, destination IP address, protocol type, and warning message related to parameter tampering detection.

The alert message confirms that Snort successfully detects abnormal HTTP traffic generated by manipulated parameters in the ARDWD web application.

This analysis shows that network-level monitoring using Snort IDS can identify suspicious traffic even when the web application fails to detect parameter tampering at the application layer.

```
analyst@secOps ~]$ sudo snort -A console -q -u snort -g snort -c /etc/snort/rules/local.rules -i enp0s3  
04/06-10:01:18.179539  [**] [1:1000001:1] WARNING: PARAMETER TAMPERING DETECTED [**] [Priority: 0] {TCP} 10.25.68.189:35978 -> 10.25.68.134:80
```

VULNERABILITY CLASSIFICATION

The identified vulnerability in the ARDWD web application is classified as Parameter Tampering, which is closely related to Broken Access Control and insufficient server-side validation. Parameter tampering occurs when attackers modify HTTP request parameters during transmission to manipulate system behavior without proper authorization or verification.

In this case, the application allows user-submitted parameters such as Customer_name, Address, and Order_item to be transmitted through HTTP POST requests without integrity verification. The server does not validate whether the submitted data has been modified during transmission, allowing attackers to inject manipulated values into the system.

This vulnerability is categorized under Broken Access Control because the application fails to enforce proper validation and integrity checks on user input, allowing unauthorized parameter modification. As a result, attackers can alter transaction data and generate abnormal network traffic that bypasses application-level security controls.

Snort IDS is used in this project to monitor network traffic and detect suspicious parameter manipulation at the network layer, providing an additional security monitoring mechanism.

SECURITY ANALYSIS

The main security weakness of the ARDWD web application lies in the absence of strong server-side validation and network monitoring mechanisms. The system accepts HTTP POST parameters without verifying their integrity, allowing attackers to manipulate transaction data during transmission.

Without proper monitoring, abnormal HTTP traffic generated by parameter tampering attacks may pass through the network undetected. This creates a security gap where application-level vulnerabilities can be exploited without triggering system warnings.

Snort Intrusion Detection System provides network-level visibility by monitoring HTTP traffic and detecting suspicious parameter values in real time. This allows security analysts to observe abnormal network behavior and identify potential attacks even when the application fails to detect manipulation.

However, Snort relies on signature-based detection and predefined rules, meaning it can only detect known suspicious patterns. Snort cannot prevent attacks or fix application vulnerabilities, and it should be used as a complementary security mechanism alongside proper server-side validation and secure coding practices.

This analysis shows that network monitoring improves system security but cannot replace secure application design.

IMPACT ANALYSIS

The Parameter tampering vulnerabilities can have significant impacts on web application security and network integrity. When attackers manipulate HTTP POST parameters, the system may process unauthorized or incorrect transaction data, leading to data integrity violations and operational risks.

Unauthorized transaction modification can allow attackers to alter order information, such as customer details or order items, which may result in fraudulent transactions or incorrect system records. This can affect business operations and reduce system reliability.

Additionally, abnormal network traffic generated by parameter tampering attacks may indicate potential intrusion attempts. If such traffic is not monitored, attackers may continue exploiting the system without detection, increasing the risk of undetected network intrusion and system compromise.

Operational disruption may also occur if manipulated data affects system processing or database integrity. Therefore, continuous network monitoring and traffic analysis are essential to detect suspicious activity and minimize security risks..

MITIGATION STRATEGIES

Several mitigation strategies can be implemented to reduce parameter tampering risks and improve web application security.

Server-side validation should be implemented to verify all incoming HTTP POST parameters and ensure that manipulated data cannot be processed by the system. Input integrity verification mechanisms such as parameter validation and data filtering can prevent unauthorized modifications.

HTTPS communication should be used to encrypt data transmission between the client and the server, reducing the risk of interception and manipulation during network communication.

Access control enforcement is also necessary to ensure that only authorized users can submit and modify transaction data. Proper authentication and authorization mechanisms can prevent unauthorized parameter changes.

Snort IDS monitoring should be continuously applied to observe network traffic and detect suspicious activity in real time. Regular network traffic analysis and security logging can help security administrators identify potential attacks early and respond quickly.

By combining application-level security and network-level monitoring, the system can achieve stronger protection against parameter tampering attacks.

CONCLUSION

This project extends the previous parameter tampering study by analyzing network traffic generated during the attack using the Snort Intrusion Detection System in the ARDWD web-based ordering application. The study focuses on observing abnormal HTTP traffic patterns created by parameter tampering and evaluating how Snort detects suspicious network activity in a controlled cybersecurity laboratory environment.

The results show that parameter tampering attacks generate abnormal HTTP traffic that can be monitored and detected through Snort IDS alert mechanisms. Snort successfully identifies manipulated parameters and provides real-time alert output containing important network information such as source IP address, destination IP address, and warning messages related to suspicious traffic. This demonstrates that network-level monitoring can improve visibility into malicious activity even when the application layer fails to detect parameter manipulation.

The study highlights the importance of combining application security and network traffic monitoring to strengthen system protection. While Snort IDS provides effective detection of abnormal traffic, it cannot replace proper server-side validation and secure coding practices. Therefore, intrusion detection systems should be implemented alongside secure communication protocols, access control mechanisms, and continuous network monitoring to ensure comprehensive web application security.

Overall, this project shows that network traffic analysis using Snort IDS plays a significant role in identifying parameter tampering attacks and enhancing cybersecurity defense mechanisms in web-based systems.

PROJECT DEVELOPER

Develop by	: Azzahra Fatma Wahidah & Nazilatun Natasya Ramadhani
Hardware	: HP Laptop 15-dy5xxx
Operating System	: 1. Microsoft Window 11 (victim) 2. Kali Linux (attacker) 3. Cybersecurity Lab VMworkstation (server and security)
Software	: 1. Microsoft word 2. Microsoft PowerPoint 3. Canva 4. Burp Suite 5. VirtualBox 6. MySQL 7. Apache 8. PHP 9. Snort